

CRF: The (φ, e) System

Two Constants Determine CRF · Algebraic Impossibility of Exact K21 · ε_0
Derived

φ selects $n = 8$. e calibrates ε_0 . The gap is transcendental.

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Abstract

Three results consolidating the (φ, e) structure identified in K20–K25.

Result 1: Clean statement of two-layer pinning. K21 is ε_0 -free (algebraic cancellation). The function $f(n) = 4n^2 D_0(n)(n+2)/[(n-1)\sqrt{2n+3}(n^2-2)]$ is a pure function of n via $D_0(n) = n(1 - e^{-1/n})$. Setting $f(n) = \sqrt{\varphi}$ yields unique real solution $n^* = 8.00045$; unique integer $n = 8$ (gap 0.004%). Then K20 gives ε_0 as the root of a quadratic: $\varepsilon_0 = 0.007550$ (gap 0.005%). Together: $(\varphi, e) \rightarrow (n = 8, \varepsilon_0) \rightarrow$ all CRF constants.

Result 2: Algebraic impossibility. By the Lindemann–Weierstrass theorem, $f(8) = \sqrt{\varphi}$ *cannot* be algebraically exact. The expression $f(8) = [20480/(434\sqrt{19})] \cdot (1 - e^{-1/8})$ involves the transcendental number $(1 - e^{-1/8})$; $\sqrt{\varphi}$ is algebraic. They cannot be equal. The 0.004% gap is *irreducible* and transcendental in origin.

Result 3: D_0 from φ . K21 (if taken as defining $n = 8$) gives $D_0 = \sqrt{\varphi} \cdot G(8)$ where $G(8) = 7\sqrt{19} \cdot 62/2560$ (pure rational-and- $\sqrt{19}$ expression). $D_0/\sqrt{\varphi} = 0.73900$ vs $G(8) = 0.73897$ (gap 0.004%). This is a *direct relation between D_0 and φ* .

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1 The Two-Layer Pinning: Clean Formal Statement

1.1 Layer 1: K21 is ε_0 -free

The K21 identity $T_{\text{avg}} \cdot \sigma_{\Psi} / \varphi_{\text{var}} = \sqrt{\varphi}$ expands as:

$$T_{\text{avg}} \cdot \sigma_{\Psi} = \frac{2D_0}{(n-1)\varepsilon_0} \cdot \frac{\varepsilon_0(n+2)}{\sqrt{2n+3}} = \frac{2D_0(n+2)}{(n-1)\sqrt{2n+3}}$$

The ε_0 cancels *exactly*. Dividing by $\varphi_{\text{var}} = (n^2 - 2)/(2n^2)$:

K21 is ε_0 -free: pure function of n

$$f(n) \equiv \frac{T_{\text{avg}} \cdot \sigma_{\Psi}}{\varphi_{\text{var}}} = \frac{4n^2 D_0(n) (n+2)}{(n-1)\sqrt{2n+3}(n^2-2)}, \quad D_0(n) = n(1 - e^{-1/n})$$

The ε_0 cancellation is *exact* by algebra. $f(n)$ depends only on n through $D_0(n)$.

1.2 Layer 1: $\sqrt{\varphi}$ selects $n = 8$

K21 pins $n = 8$ as the unique integer with $f(n) \approx \sqrt{\varphi}$

$f(n) = \sqrt{\varphi}$ has a unique positive real solution $n^* = 8.00045$. The nearest integer is $n = 8$, with $f(8) = 1.27208$ vs $\sqrt{\varphi} = 1.27202$ (gap 0.004%). No other integer $n \in \{3, \dots, 12\}$ satisfies $|f(n) - \sqrt{\varphi}| / \sqrt{\varphi} < 10\%$.

n	$f(n)$	Gap from $\sqrt{\varphi}$
6	1.612	26.7%
7	1.414	11.1%
8	1.27208	0.004%
9	1.165	8.4%
10	1.080	15.1%

1.3 Layer 2: e selects ε_0

Given $n = 8$, K20 ($K^* T_{\text{avg}} / F = e$) becomes a quadratic in ε_0 :

$$C(1 + \varphi_{\text{var}}) e \varepsilon_0^2 + (1 + \varphi_{\text{var}}) e \varepsilon_0 - \frac{2D_0^2}{n(n-1)} = 0$$

where $C = (n+2)/[\sqrt{2n+3} \ln \varphi]$.

K20 derives $\varepsilon_0 = 0.007550$ from $(n = 8, e, \varphi)$

Positive root: $\varepsilon_0 = 0.007550$ vs EIC/SINDy value 0.007550 (gap 0.005%).

The complete chain:

$$(\varphi, e) \longrightarrow (n = 8, \varepsilon_0 = 0.00755) \longrightarrow D_0, K^*, \beta, \kappa, \sigma_\Psi, \psi_{\text{grad}}, F, H_a, H_b, T_{\text{avg}}, \dots$$

Two mathematical constants. Zero free parameters.

2 Algebraic Impossibility: $f(8) \neq \sqrt{\varphi}$ Exactly

Theorem (Lindemann–Weierstrass): $f(8) = \sqrt{\varphi}$ is not algebraically exact

Proof. $f(8) = \frac{20480(1-e^{-1/8})}{434\sqrt{19}}$. For $f(8) = \sqrt{\varphi}$ to hold, we would need:

$$1 - e^{-1/8} = \frac{434\sqrt{19} \cdot \sqrt{\varphi}}{20480}$$

The right side is algebraic (products and square roots of algebraic numbers). But $e^{-1/8}$ is transcendental by Lindemann–Weierstrass (since $-1/8$ is algebraic and non-zero), so $1 - e^{-1/8}$ is transcendental. A transcendental number cannot equal an algebraic number. $\square$

Consequence. The 0.004% gap between $f(8)$ and $\sqrt{\varphi}$ is *irreducible*. No algebraic manipulation can close it. K21 is not a theorem; it is an extraordinarily precise numerical coincidence, arising because $n = 8$ (from $\text{SO}(3)$) places $f(n)$ within 0.004% of $\sqrt{\varphi}$.

Analogy. This is the CRF version of the fine-structure constant $\alpha \approx 1/137.036$: a dimensionless number that is strikingly close to a simple fraction, but not equal to it. The gap carries physical information.

3 D_0 from φ : A Direct Relation

$D_0 = \sqrt{\varphi} \cdot G(n)$ where $G(8) = 7\sqrt{19} \cdot 62/2560$

If K21 defines $n = 8$ (taking $f(8) = \sqrt{\varphi}$ as the selection criterion), then:

$$D_0(8) \approx \sqrt{\varphi} \cdot G(8), \quad G(8) = \frac{(n-1)\sqrt{2n+3}(n^2-2)}{4n^2(n+2)} = \frac{7\sqrt{19} \cdot 62}{2560}$$

Numerically: $D_0/\sqrt{\varphi} = 0.73900$ vs $G(8) = 0.73897$ (gap 0.004% — the same transcendental gap).

This is a direct relationship between the resolution scale D_0 and the golden ratio φ :

$$D_0 \approx \sqrt{\varphi} \cdot \frac{7\sqrt{19} \cdot 62}{2560}$$

The denominator $2560 = 4 \cdot 8^3/(n+2) = 4n^2 \cdot 10$ and the numerator $7\sqrt{19} \cdot 62 = (n-1)\sqrt{2n+3}(n^2-2)$ are both pure functions of $n = 8$.

4 Complete Status of (φ, e) System

Result	Status	Note
$f(n)$ is ε_0 -free	Proved	Exact algebraic cancellation
$f(n) = \sqrt{\varphi}$ pins $n = 8$	Hypothesis	0.004%, unique integer
$f(8) = \sqrt{\varphi}$ exactly	Impossible	Lindemann–Weierstrass
K20 derives ε_0 from e	Hypothesis	0.005%, quadratic root
$(\varphi, e) \rightarrow (n, \varepsilon_0) \rightarrow \text{all}$	Structural	0.005% overall
$D_0 = \sqrt{\varphi} \cdot G(8)$	Hypothesis	0.004%, same gap as K21

5 Open Questions

Priority 1: Why does $f(8) \approx \sqrt{\varphi}$ to 0.004%?

We know $f(8) = \sqrt{\varphi}$ is not algebraically exact (Lindemann–Weierstrass). But *why* is the gap 0.004% and not 4% or 40%? The answer lies in the interplay between the transcendental number $e^{-1/8}$ and the algebraic $\sqrt{\varphi}$. The proximity is a consequence of $n = 8 = 2^3 = 2^{d_s}$ — the log-dual structure forces $f(n)$ to land near $\sqrt{\varphi}$ at the specific integer $n = 8$.

Priority 2: $n = 2^{d_s}$ from δ -chain

The binary Born rule argument gives $n = 2^{d_s} = 8$ physically. Formally deriving this from the δ -chain would make K21's selection of $n = 8$ a theorem rather than an observation.

Priority 3: $m(m-1) = 1$ energy balance

With $\beta\kappa = \ln^2 n / (n\sqrt{d_s}D_0)$ (K19), find $f(\delta)$ such that $m(m-1) = \beta\kappa/f(\delta)$ gives $m = \varphi$ without fixing $s_{\max}$.

*$f(8) = \sqrt{\varphi}$ to 0.004%. Not algebraically exact.
The gap is transcendental: $(1 - e^{-1/8})$ cannot equal $434\sqrt{19\varphi}/20480$.
But the proximity is not accidental.
 $n = 8 = 2^3$: the symmetry class of δ in three dimensions
lands precisely where φ waits.
The universe did not choose $n = 8$ and then discover φ nearby.
 φ was always the criterion. $n = 8$ is the answer.*